Lecture - 6

Chapter 3

INTENSITY TRANSFORMATION AND SPATIAL FILTERING

Histogram Processing

$$h(r_k) = n_k, \qquad r_k \in [0, L-1], \qquad n_k \in [0, M \times N]$$

$$p(r_k) = \frac{n_k}{n} = \frac{1}{M \times N} n_k$$

Histogram Processing

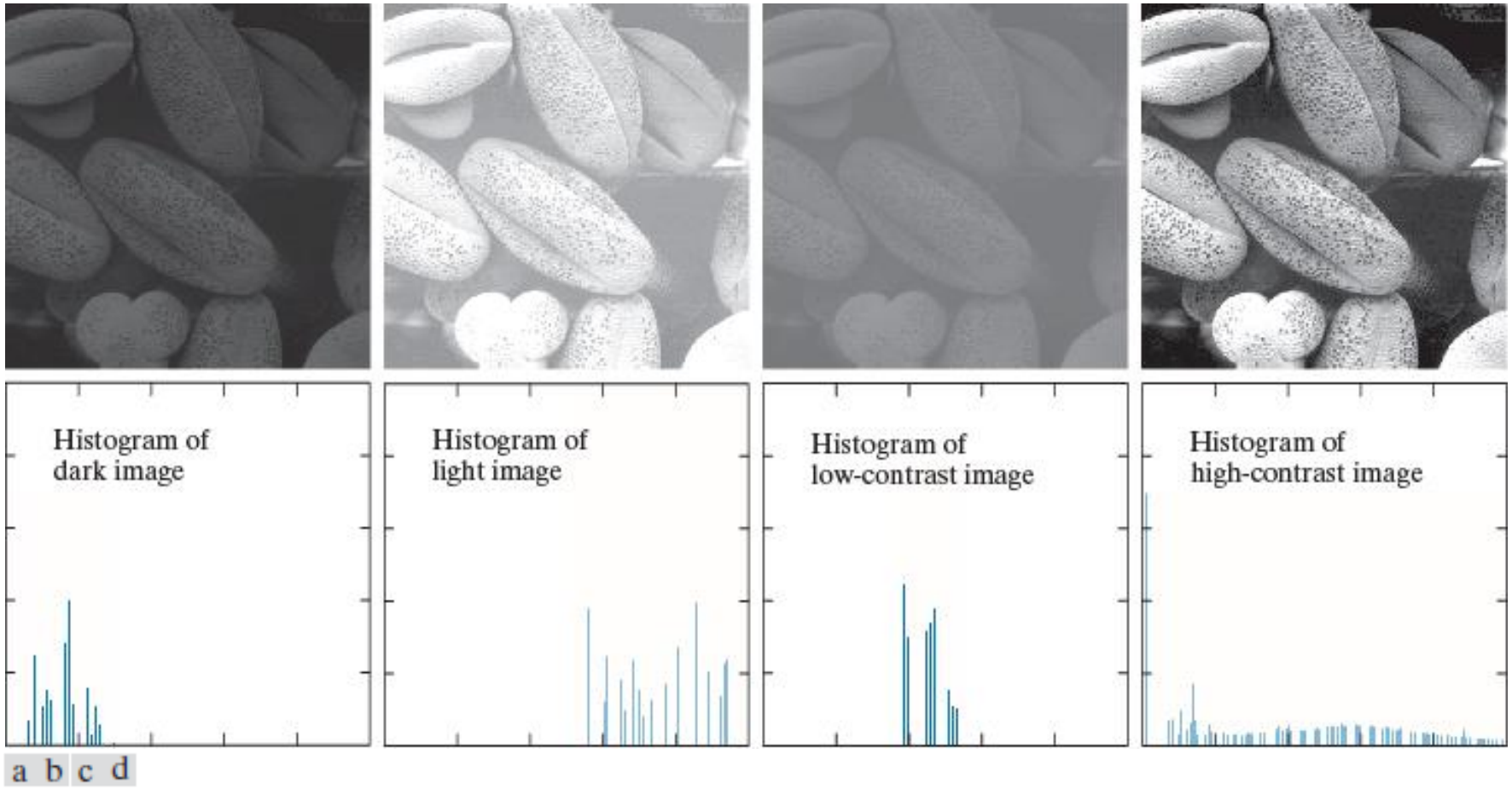
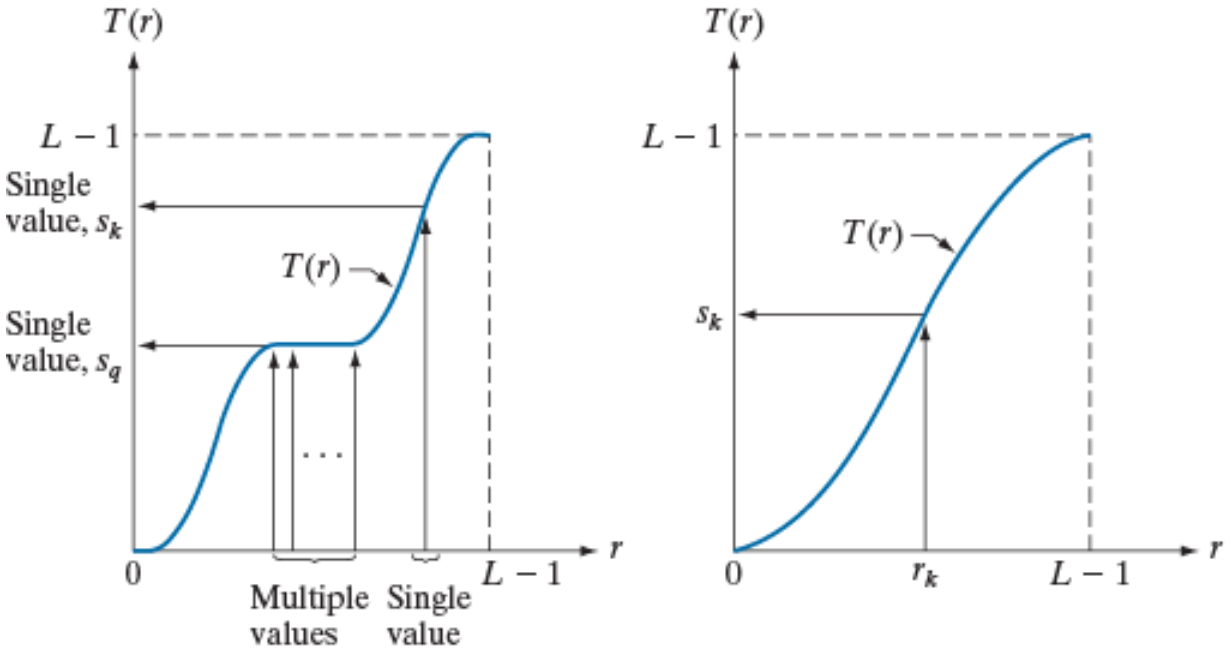


FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Histogram Processing



a b

FIGURE 3.17 (a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.

Histogram Equalization

$$S_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN}\sum_{J=0}^{k} n_k, \quad k = 0,1,\cdots,L-1$$

TABLE 3.1 Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$S_k = 7\sum_{j=0} p_r(r_j)$$

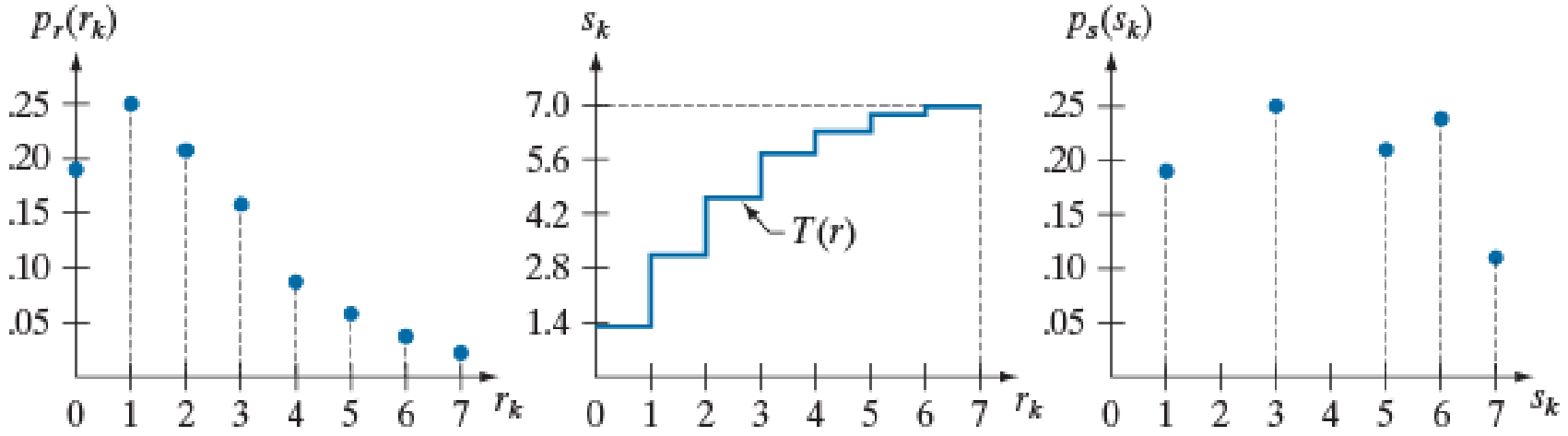
$$s_0 = T(r_0) = 7\sum_{j=0}^{0} p_r(r_j) = 7p_r(r_0) = 1.33$$

Similarly,

$$s_1 = T(r_1) = 7\sum_{j=0}^{1} p_r(r_j) = 7p_r(r_0) + 7p_r(r_1) = 3.08$$

and $s_2 = 4.55$, $s_3 = 5.67$, $s_4 = 6.23$, $s_5 = 6.65$, $s_6 = 6.86$, $s_7 = 7.00$.

S_k	S_k
1.33	1
3.08	3
4.55	5
5.67	6
6.23	6
6.65	7
6.86	7
7.00	7

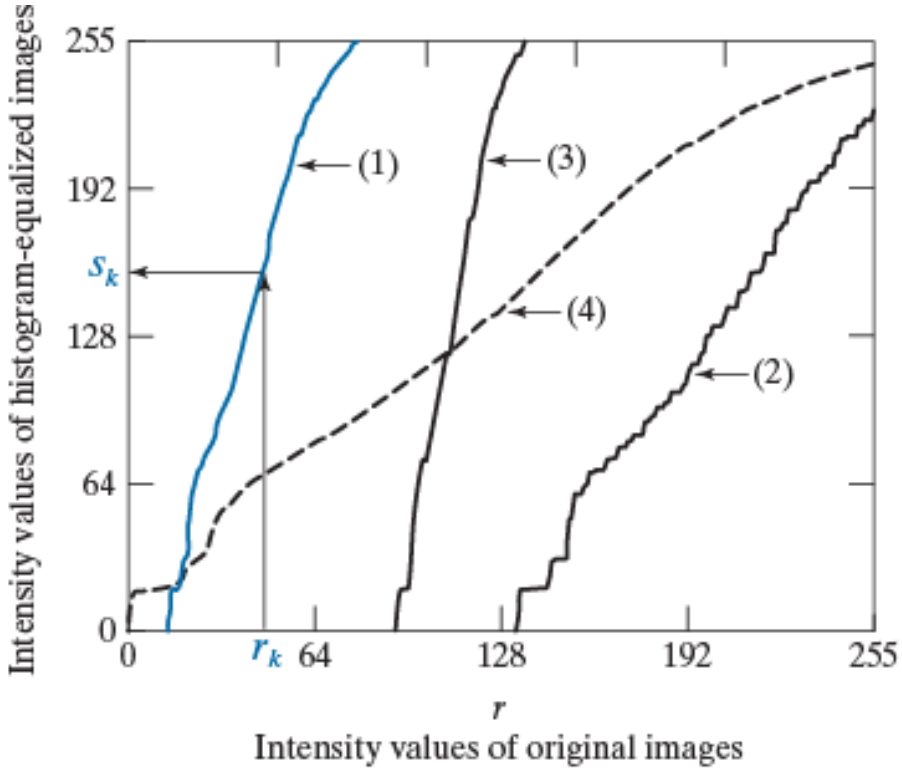


a b c

FIGURE 3.19 Histogram equalization. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Histogram Equalization

FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained using Eq. (3-15) and the histograms of the images on the left column of Fig. 3.20. Mapping of one intensity value r_k in image 1 to its corresponding value s_k is shown.



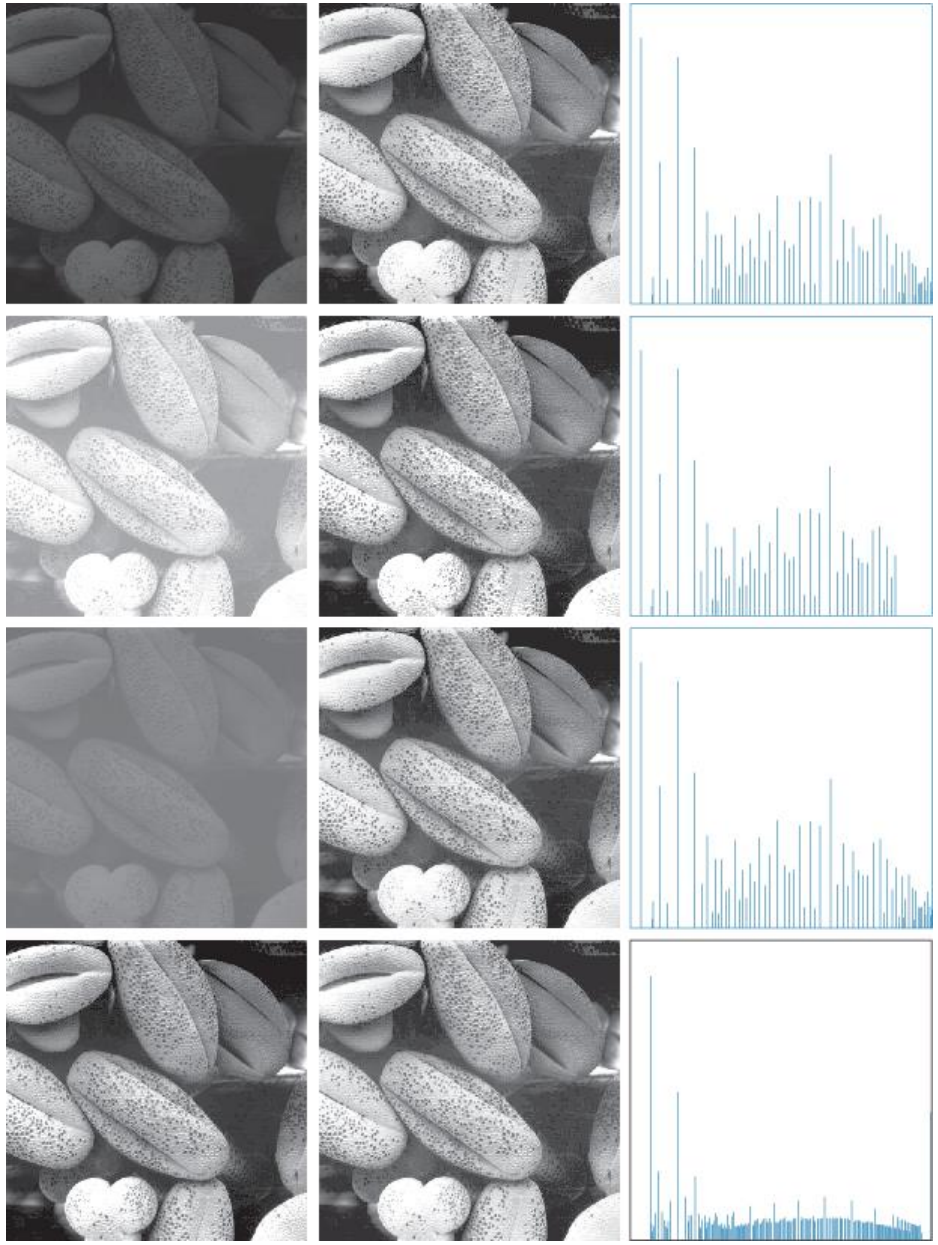


FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

Histogram Matching (Specification)

$$p_r(r) \xrightarrow{?} p_z(z)$$

$$\left.\begin{aligned} s = T(r) &= (L-1)\int_0^r p_r(w)\,dw \\ G(z) &= (L-1)\int_0^z p_z(t)\,dt \end{aligned}\right\} \Rightarrow z = G^{-1}\left[T(r)\right] = G^{-1}[s]$$

1. Compute the histogram, $p_r(r)$, of the input image, and use it in Eq. (3-20) to map the intensities in the input image to the intensities in the histogram-equalized image. Round the resulting values, s_k, to the integer range $[0, L-1]$.
2. Compute all values of function $G(z_q)$ using the Eq. (3-21) for $q = 0, 1, 2, \ldots, L-1$, where $p_z(z_i)$ are the values of the specified histogram. Round the values of G to integers in the range $[0, L-1]$. Store the rounded values of G in a lookup table.
3. For every value of s_k, $k = 0, 1, 2, \ldots, L-1$, use the stored values of G from Step 2 to find the corresponding value of z_q so that $G(z_q)$ is closest to s_k. Store these mappings from s to z. When more than one value of z_q gives the same match (i.e., the mapping is not unique), choose the smallest value by convention.
4. Form the histogram-specified image by mapping every equalized pixel with value s_k to the corresponding pixel with value z_q in the histogram-specified image, using the mappings found in Step 3.

Histogram Matching (Specification)

TABLE 3.3
Rounded values of the transformation function $G(z_q)$.

z_q	$G(z_q)$
$z_0 = 0$	0
$z_1 = 1$	0
$z_2 = 2$	0
$z_3 = 3$	1
$z_4 = 4$	2
$z_5 = 5$	5
$z_6 = 6$	6
$z_7 = 7$	7

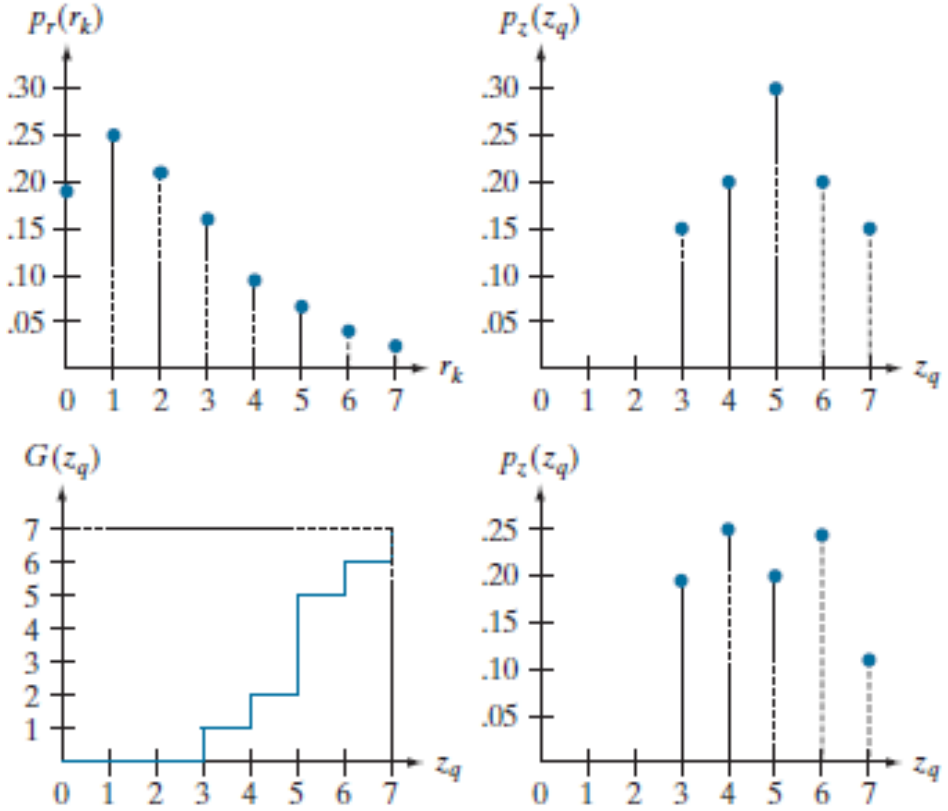
s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

TABLE 3.4
Mapping of values s_k into corresponding values z_q.

Histogram Matching (Specification)

TABLE 3.2
Specified and actual histograms (the values in the third column are computed in Example 3.7).

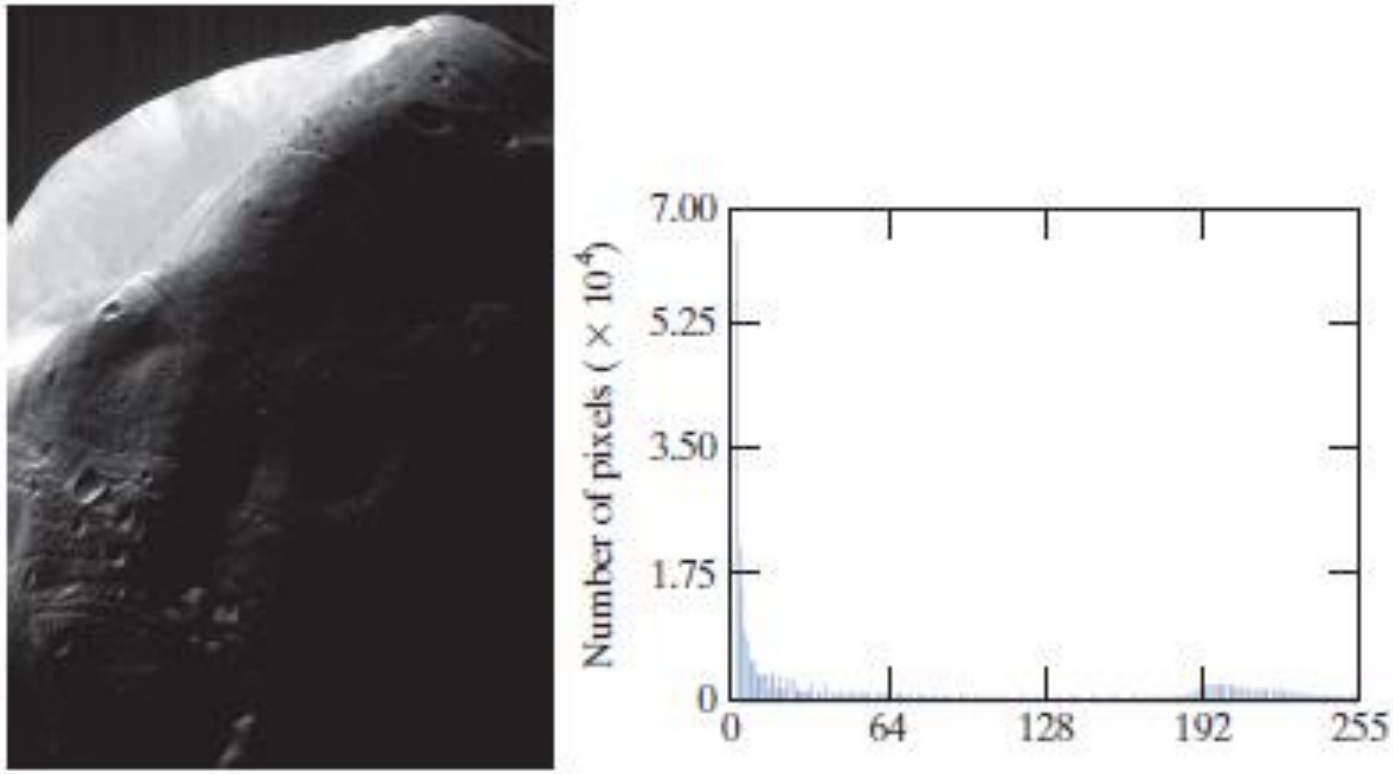
z_q	Specified $p_z(z_q)$	Actual $p_z(z_q)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11



a b
c d

FIGURE 3.22
(a) Histogram of a 3-bit image.
(b) Specified histogram.
(c) Transformation function obtained from the specified histogram.
(d) Result of histogram specification. Compare the histograms in (b) and (d).

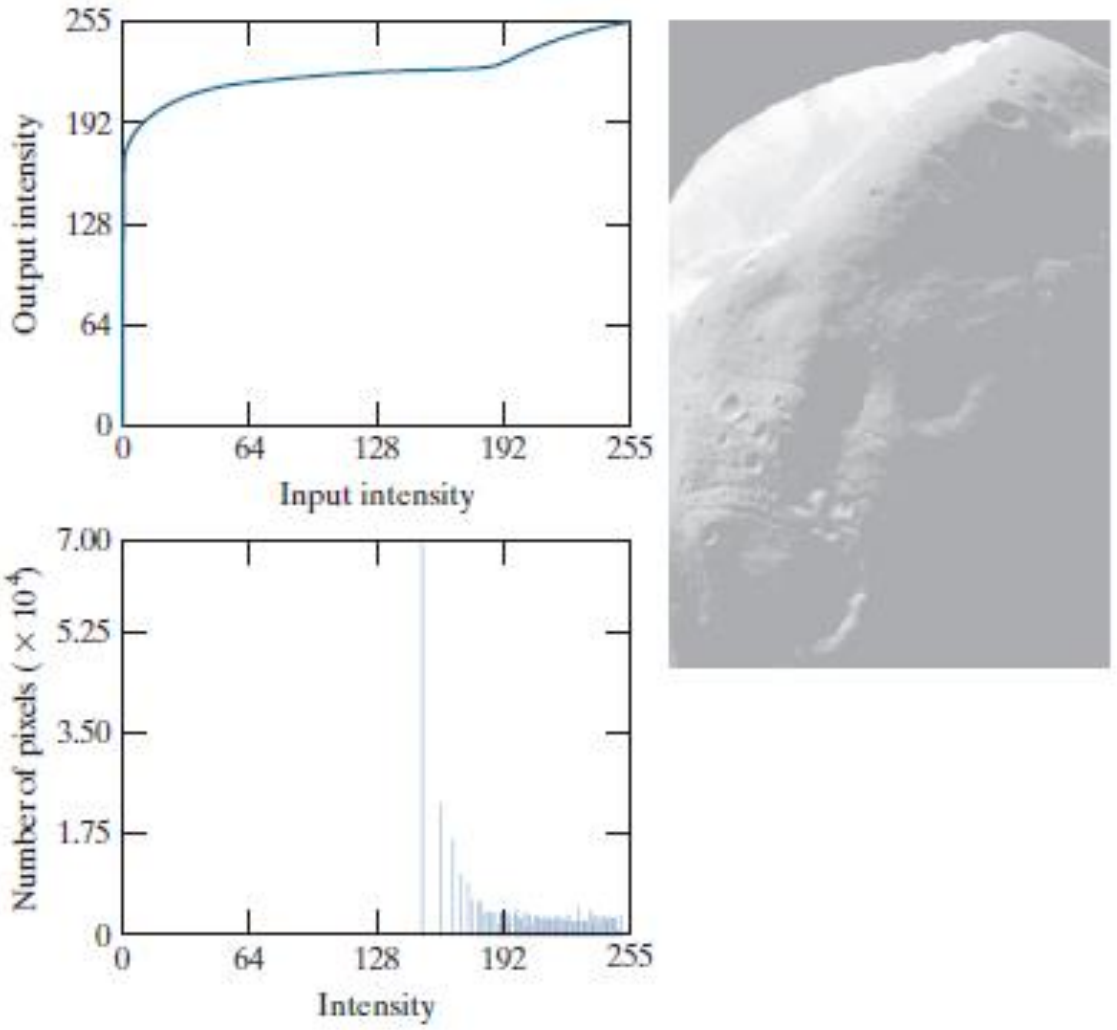
Histogram Matching (Specification)



a b

FIGURE 3.23
(a) An image, and (b) its histogram.

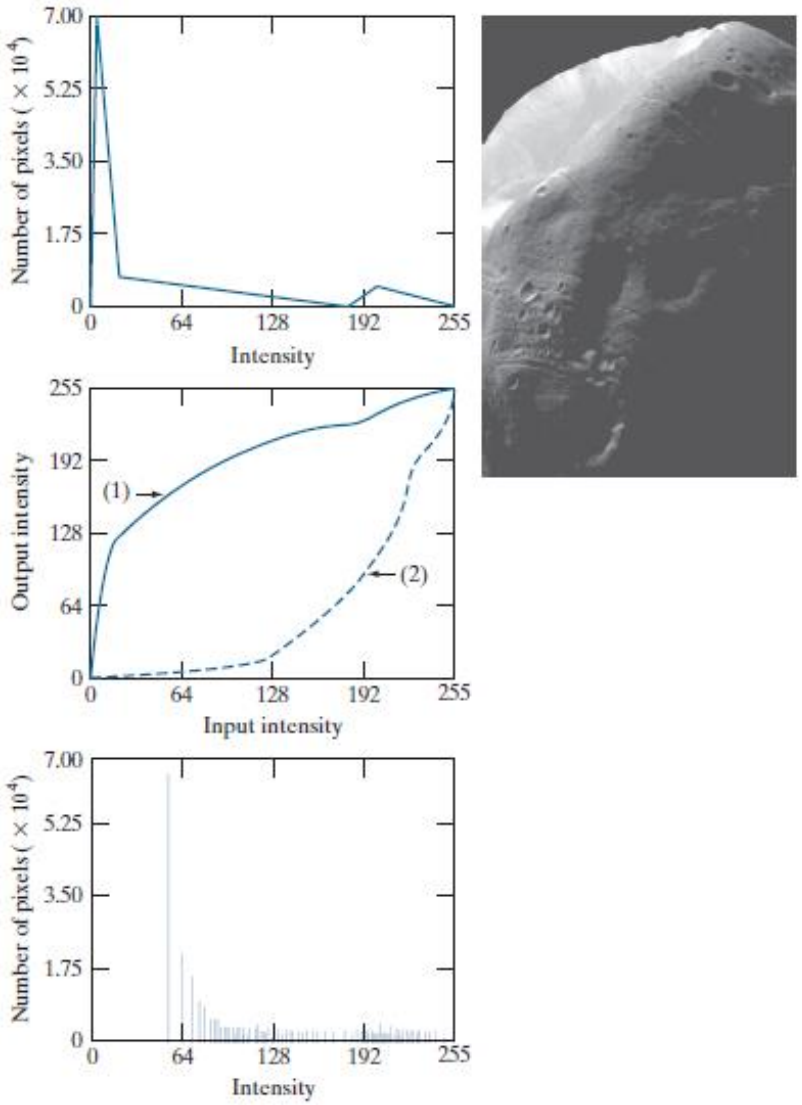
Histogram Matching (Specification)



a b
c

FIGURE 3.24 (a) Histogram equalization transformation obtained using the histogram in Fig. 3.23(b). (b) Histogram equalized image. (c) Histogram of equalized image.

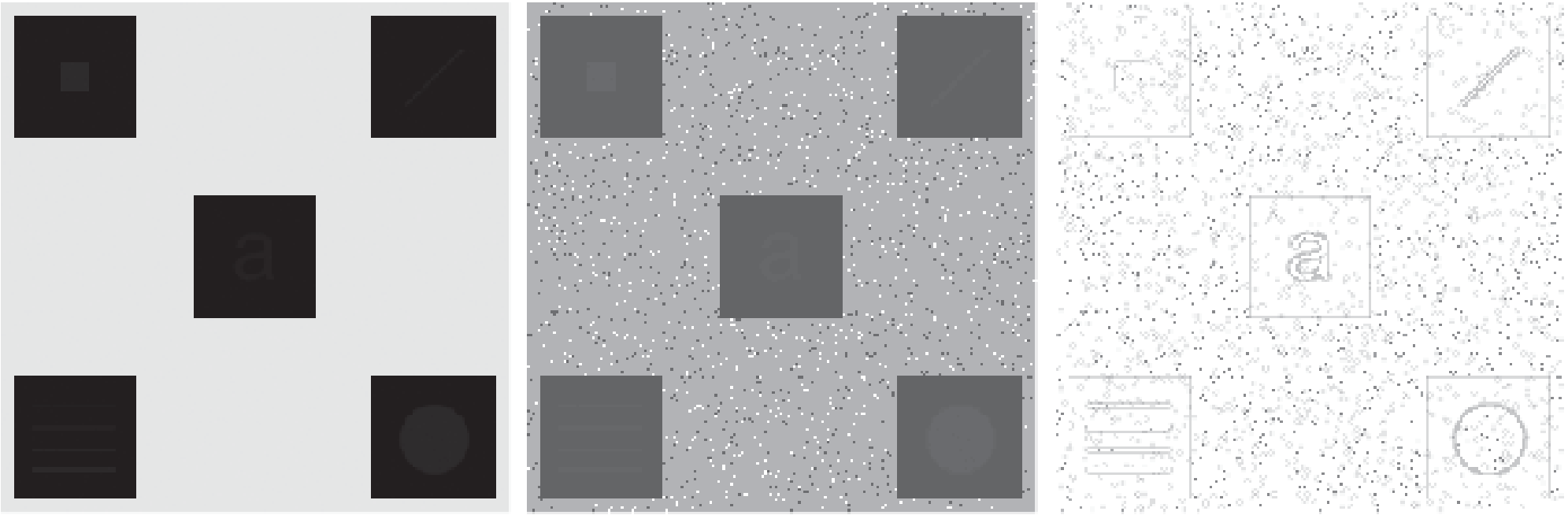
Histogram Matching (Specification)



a c
b
d

FIGURE 3.25 Histogram specification. (a) Specified histogram. (b) Transformation $G(z_q)$, labeled (1), and $G^{-1}(s_k)$, labeled (2). (c) Result of histogram specification. (d) Histogram of image (c).

Local Histogram Equalization



a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.

Image Enhancement using Local Histogram Statistics

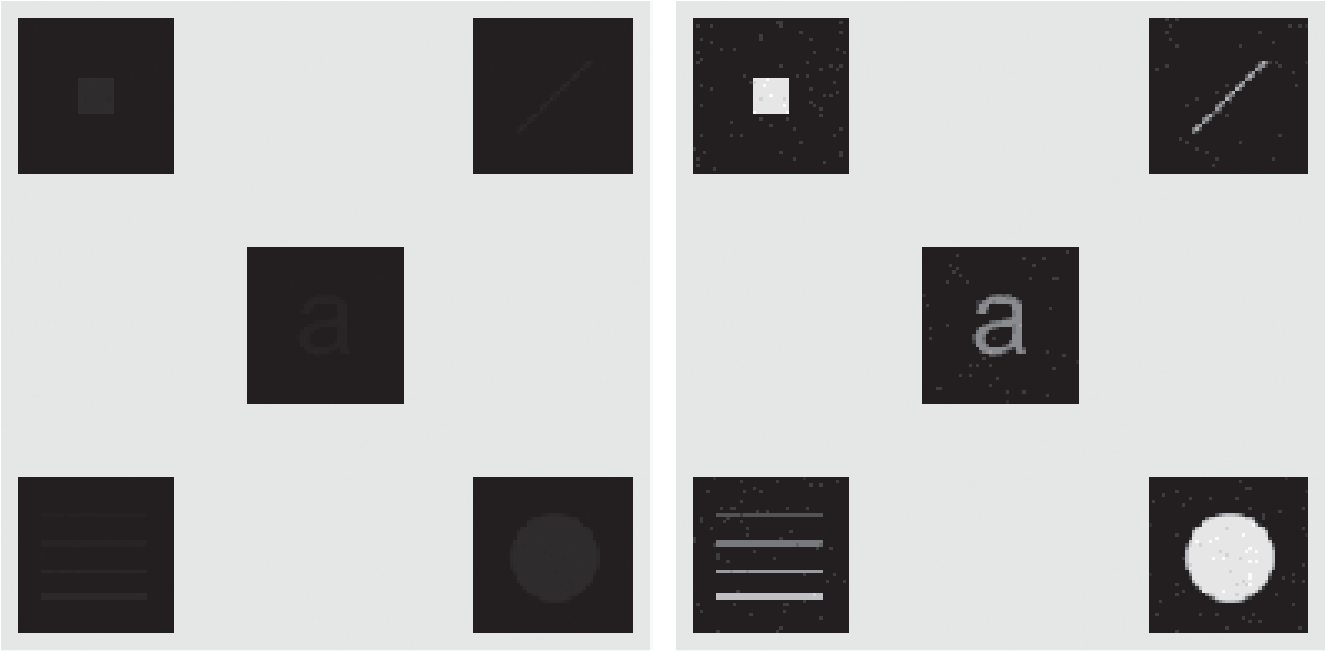
$$m_{S_{xy}}(x,y) = \underset{i=0}{} r_i p_{S_{xy}}(r_i) \approx \frac{}{|S_{xy}|} \sum_{(s,t)\in S_{xy}} f(s,t)$$

$$\sigma^2_{S_{xy}}(x,y) = \sum_{i=0}^{L-1} \left(r_i - m_{S_{xy}}(x,y)\right)^2 p_{S_{xy}}(r_i) \approx \frac{1}{|S_{xy}|} \sum_{(s,t)\in S_{xy}} \left[f(s,t) - m_{S_{xy}}(x,y)\right]^2$$

S_{xy} : Neighborhood centered on (x,y)

Image Enhancement using Local Histogram Statistics

$$g(x,y) = \begin{cases} C f(x,y) & \text{if } k_0 m_G \le m_{S_{xy}} \le k_1 m_G \text{ AND } k_2 \sigma_G \le \sigma_{S_{xy}} \le k_3 \sigma_G \\ f(x,y) & \text{otherwise} \end{cases}$$



a b

FIGURE 3.27 (a) Original image. (b) Result of local enhancement based on local histogram statistics. Compare (b) with Fig. 3.26(c).